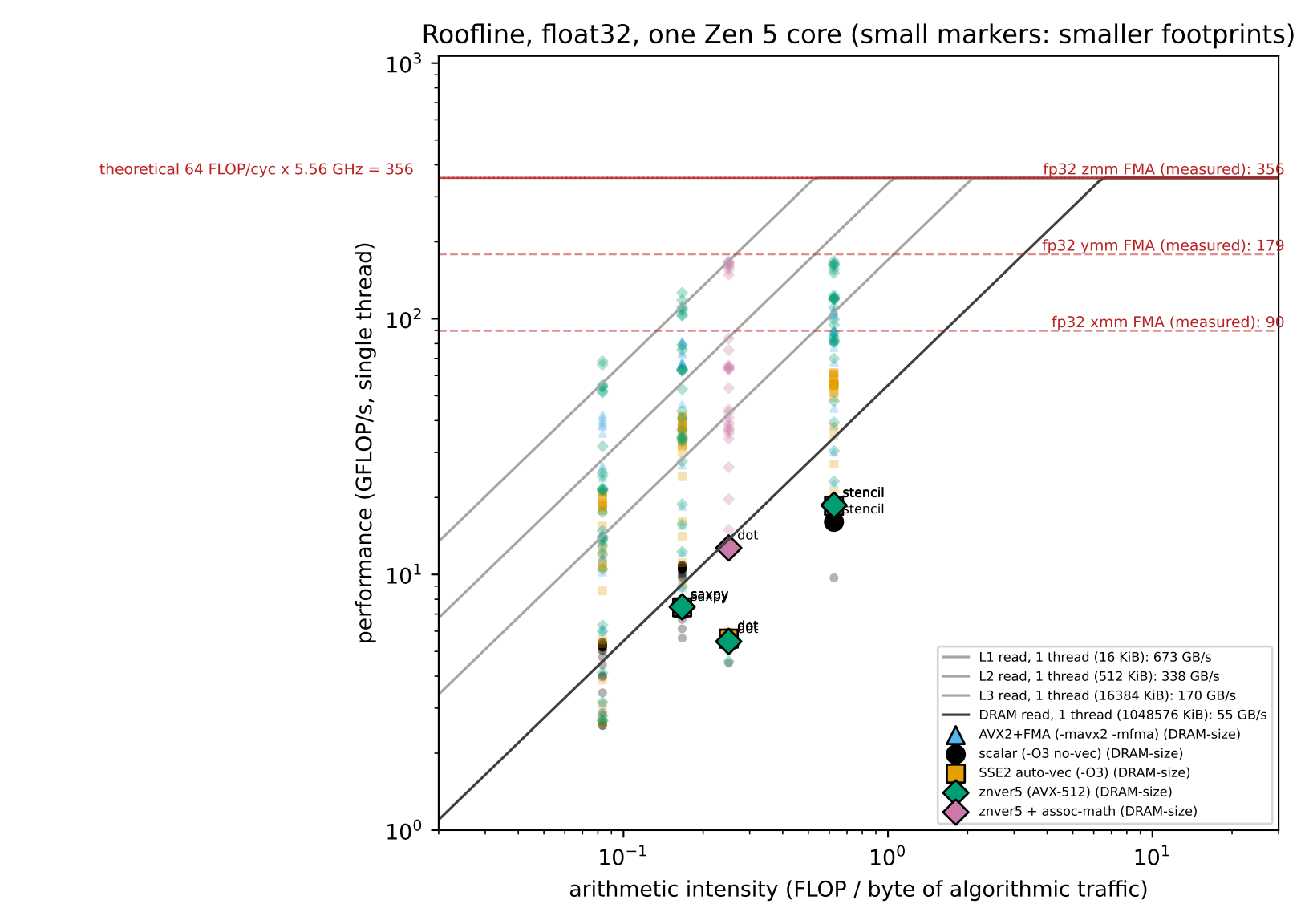
Roofline, float32, one Zen 5 core (small markers: smaller footprints)
theoretical 64 FLOP/cyc x 5.56 GHz = 356
fp32 zmm FMA (measured): 356
fp32 ymm FMA (measured): 179
fp32 xmm FMA (measured): 90
performance (GFLOP/s, single thread)
arithmetic intensity (FLOP / byte of algorithmic traffic)
stencil
stencil
dot
saxpy
saxpy
dot
dot
L1 read, 1 thread (16 KiB): 673 GB/s
L2 read, 1 thread (512 KiB): 338 GB/s
L3 read, 1 thread (16384 KiB): 170 GB/s
DRAM read, 1 thread (1048576 KiB): 55 GB/s
AVX2+FMA (-mavx2 -mfma) (DRAM-size)
scalar (-O3 no-vec) (DRAM-size)
SSE2 auto-vec (-O3) (DRAM-size)
znver5 (AVX-512) (DRAM-size)
znver5 + assoc-math (DRAM-size)